

EXPERIMENT 3

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Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 3: Working of Hub, Switch, and Router

Aim

To examine the functions of a Hub, Switch, and Router by designing different network layouts in Cisco Packet Tracer and studying how data travels between connected computers.

Objectives

- Understand the purpose and operation of Hub, Switch, and Router in computer networks.
- Build simple network topologies using Cisco Packet Tracer.
- Configure IP addresses and gateway settings for successful communication.
- Observe the movement of packets in simulation mode and compare the behavior of different networking devices.

Theory

Hub

A Hub is a simple networking device that operates at the **Physical Layer (Layer 1)** of the OSI model. It connects multiple computers within a Local Area Network (LAN). Whenever a data frame reaches one port, the Hub sends the same data to every other connected port without checking the intended destination. As a result, all connected devices receive the transmitted data, making communication less efficient. Hubs are commonly used in star topology networks and do not maintain any address table or memory.

Advantages

- Low-cost networking device.
- Suitable for very small networks where traffic is limited.

Limitations

- Broadcasts data to every connected device.
- Does not store any addressing information.
- Usually supports only half-duplex communication, which lowers performance.

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Switch

A Switch is an intelligent networking device that functions at the **Data Link Layer (Layer 2)** of the OSI model. It connects devices within the same LAN while maintaining a **MAC Address Table** in memory. By checking this table, the switch forwards data only to the correct destination port instead of broadcasting it to every device. This improves network speed, reduces unnecessary traffic, and increases communication efficiency.

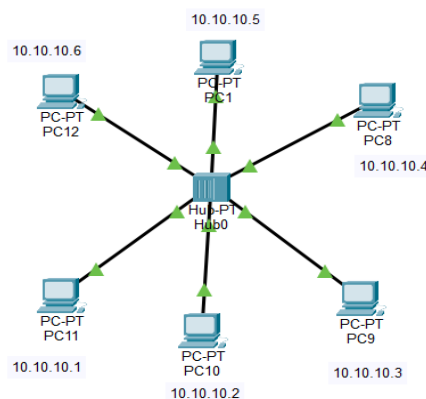
Router

A Router is a **Layer 3 (Network Layer)** device responsible for connecting two or more separate networks. It forwards packets based on IP addresses and stores routing information in a **Routing Table**. Routers enable communication between different LANs or between a LAN and a Wide Area Network (WAN). They determine the best path for data to travel from the source network to the destination network.

Practical / Experiment Steps

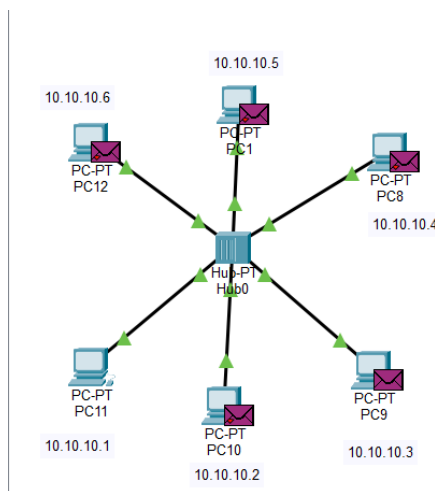
HUB Configuration

1. Hub section -> choose one hub -> click on hub and check for the ports
2. Take 5 PC's and take straight through cable -> Connect all PC's with Hub with FastEthernet port as shown in the pic 1.1



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3. On bottom right corner click on simulation -> take one data packet -> take 10.10.10.1 as Source_ip & 10.10.10.3 as Destination_ip however the packet will be broadcasted to other as well as shown in the pic 1.2.



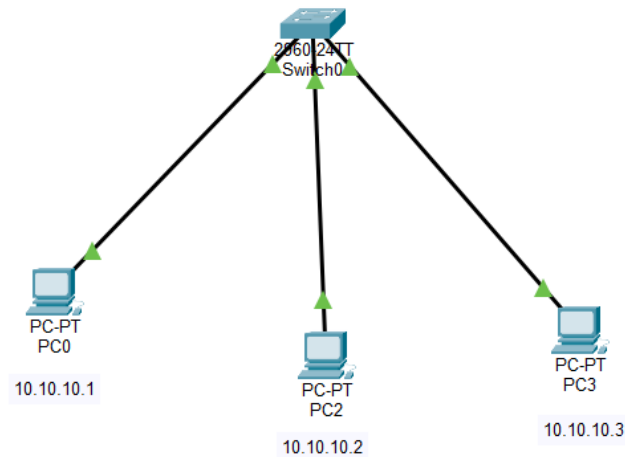
[pic 1.2]

Switches Configuration:

Steps:

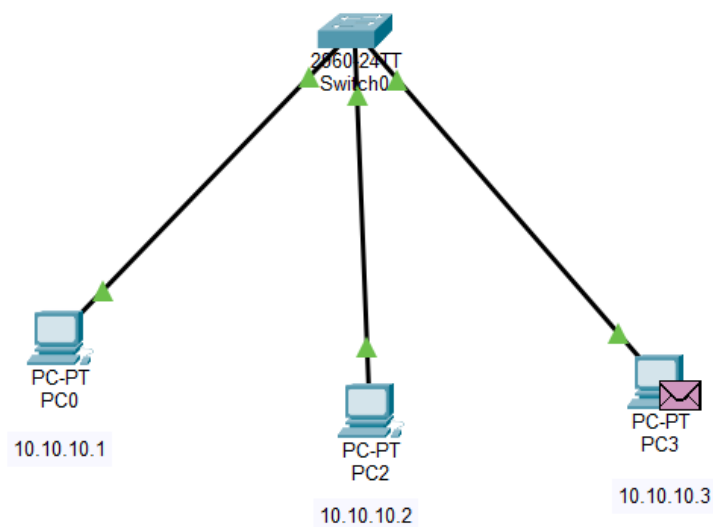
1. Switch section > choose one switch > click on switch and check for ports
2. Take 3 PC's and take straight through cable -> Connect all PC's with Switch with FastEthernet port as shown in the pic 2.1

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[pic 2.1]

3. On bottom right corner click on simulation -> take one data packet -> take 10.10.10.1 as Source_ip & 10.10.10.3 as Destination_ip and unlike hub it will not be broadcasted to all the other PCs and only be send to destination PC like shown in the pic 2.2



[pic 2.1]

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Router Configuration:

1. Configure LAN 1

- Place one 2960 switch and connect PC4 and PC5 using Copper Straight-Through cables.
- Connect the switch to Router1 using a Copper Straight-Through cable.
- Configure the IP addresses as:
 - PC4: 10.10.10.1
 - PC5: 10.10.10.2
 - Router (GigabitEthernet0/0): 10.10.10.3
- Set the subnet mask 255.255.255.0 for all devices in LAN 1.

2. Configure LAN 2

- Place another 2960 switch and connect PC6 and PC7 using Copper Straight-Through cables.
- Connect the switch to Router1 using a Copper Straight-Through cable.
- Configure the IP addresses as:
 - PC6: 192.168.1.1
 - PC7: 192.168.1.2
 - Router (GigabitEthernet0/1): 192.168.1.3
- Set the subnet mask 255.255.255.0 for all devices in LAN 2.

3. Configure the Router

- Open Router1 and assign the following IP addresses:
 - GigabitEthernet0/0: 10.10.10.3 /24
 - GigabitEthernet0/1: 192.168.1.3 /24
- Enable both interfaces using the `no shutdown` command.

4. Configure the Default Gateway

Configure the default gateway on each PC as follows:

LAN 1

- PC4 → 10.10.10.3
- PC5 → 10.10.10.3

LAN 2

- PC6 → 192.168.1.3
- PC7 → 192.168.1.3

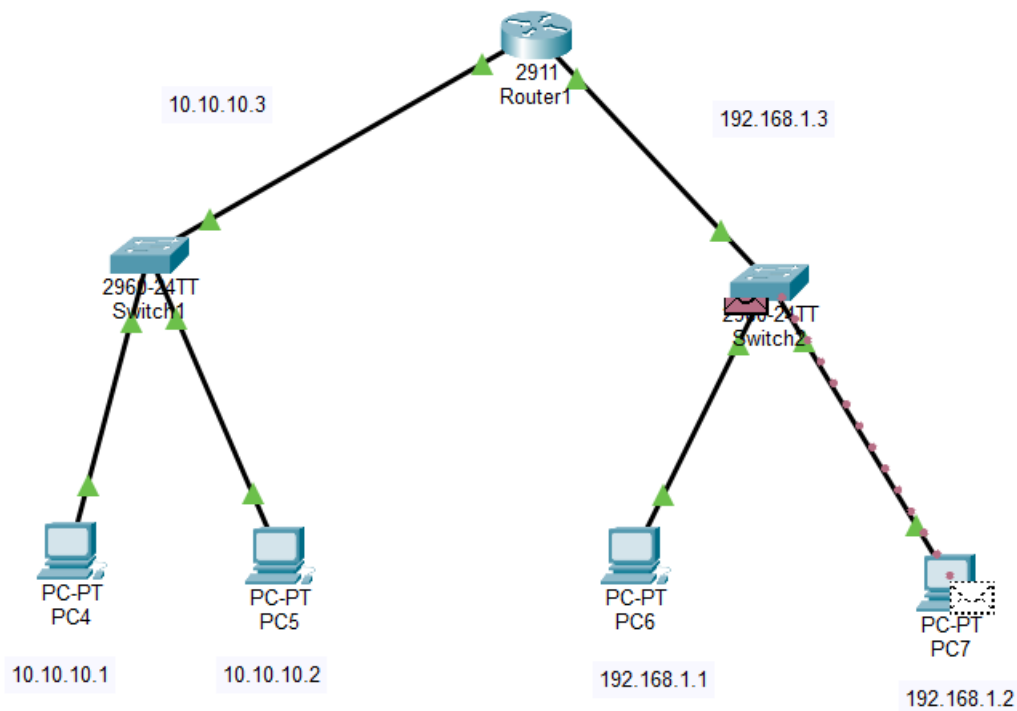
To configure the gateway:

- Open the PC.
- Go to Desktop → IP Configuration.
- Enter the Default Gateway address.

5. Test the Network

- Verify that the packet travels from **PC4 → Switch1 → Router1 → Switch2 → PC7**.

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[pic 3.1]

Learning Outcomes

After performing this experiment, students will be able to:

- Explain the roles and differences between a Hub, Switch, and Router.
- Design and simulate basic LAN and inter-network connections using Cisco Packet Tracer.
- Configure IP addressing and default gateways for network communication.
- Interpret packet flow in simulation mode and evaluate how different networking devices manage data transmission.